

Derrick Wigglesworth

derrick.wigglesworth@gmail.com | 301.802.1431 | www.drwiggle.com

SKILLS

PROGRAMMING

ChatGPT Copilot
Python Jupyter
SQL sklearn Lua
numpy pandas
bitbucket git
LaTeX powershell

ML ENGINEERING

mlflow EKS
Sagemaker Databricks

ML OPS

Database Design Docker
Kubernetes ArgoCD
Octopus CircleCI
Splunk Observe
FastApi falcon

LINKS

Github:// [drwiggle](#)
LinkedIn:// [drwiggle](#)

EDUCATION

PHD, MATHEMATICS

UNIVERSITY OF UTAH

May 2018 | Salt Lake City, UT

Thesis: [The geometry of \$Out\(F_n\)\$ through completely split train tracks](#)

BS, MATHEMATICS

BS, PHYSICS

UNIVERSITY OF MARYLAND

May 2012 | College Park, MD

Magna Cum Laude

SENIOR MACHINE LEARNING ENGINEER | PROGRESSIVE LEASING

Jan 2025 – present | Remote (CO)

- Accelerated time-to-production for ML models from 3–6 months to under 1 week by adopting productionization-first mindset: building high-quality data sets, aligning development & production environments, and validating early & often.
- Secured buy-in from multiple Directors to decouple business logic from ML model outputs, enabling rapid model iteration, reducing cross-team friction, and decreasing wasted effort.
- Architected database schemas, stored procedures, tables, etc. to meet unrecognized business needs, facilitating and simplifying analysis tasks.
- Led team effort to systematize and document model-building processes, improving model iterability and replicability.
- Built and maintained a Python package used across data science team, enabling ICs to perform analysis tasks more quickly and with a higher degree of accuracy.

DATA SCIENTIST II | PROGRESSIVE LEASING

Jan 2024 – Dec 2024 | Remote (CO)

- Built new ML model to quantify fraud risk on \$200 M in credit applications annually, increasing profits by reducing exposure to losses.
- Provided on-time delivery under tight deadlines in large effort to automate manual processes, reducing friction for the customer, and generating annual cost savings in excess of \$2 M.
- Collaborated with technical teams to implement an updated system for rapid deployment of fraud rules, while creating robust documentation and training team members in new processes.
- Saved the entire DS team an average of 3 hours per day by identifying time-sinks and architecting solutions (10 min/person * 20 people).

DATA SCIENTIST I | PROGRESSIVE LEASING

Jan 2022 – Jan 2024 | Remote (CO)

- Quickly responded to fraud attacks with error-free analyses, accurate proposals for mitigation measures, and rapid implementation.
- Worked with tech and database teams to implement an initiative to reduce business costs by \$0.5 M annually.
- Spearheaded effort to modernize tracking of flagged fraud, enabling multiple ML models and increasing the accuracy & repeatability of analyses.
- Built scripts and stored procedures to generate monthly reports on fraud attacks, losses, and mitigation measures reducing monthly effort required by 2-3 days.

VISITING ASSISTANT PROFESSOR | UNIVERSITY OF ARKANSAS

Jan 2019 – Aug 2021 | Fayetteville, AR

- **Research:** Collaborated with colleagues to advance the field of geometric group theory, publish papers in high quality journals, and shared results at national and international conferences.
- **Teaching:** Taught 2–3 classes each semester with minimal supervision:
 - Employed novel pedagogical tools (e.g., Python projects, standards based grading, flipped classroom) to enhance student experience and learning.
 - Cultivated relationships with students in class and office hours.
 - Managed deadlines when preparing and grading course materials.